

Lesson 4 · Homework

NCEA Level 2 · AS 91262 · Anti-differentiation, rates and kinematics

Time: about 60 minutes

Answer version · tutor's working, not an official mark scheme

Show all working, including $+c$. Give units and meaning for every rate. [A] Achieved, [M] Merit, [E] Excellence.

1. Anti-differentiate.

[A]

(a) $f'(x) = 8x^3$

$$f(x) = 2x^4 + c$$

(b) $f'(x) = 3x^2 - 10x + 1$

$$f(x) = x^3 - 5x^2 + x + c$$

(c) $\frac{dy}{dx} = 5 - 2x$

$$y = 5x - x^2 + c$$

(d) $f'(x) = x^2 + 4x$

$$f(x) = \frac{1}{3}x^3 + 2x^2 + c$$

2. $f'(x) = 2x + 3$ and $f(0) = -1$. Find $f(x)$.

[A]

$$f(x) = x^2 + 3x + c; f(0) = c = -1$$

$$f(x) = x^2 + 3x - 1$$

3. $f'(x) = 3x^2 - 4$ and the curve passes through (2, 5). Find $f(x)$.

[A]

$$f(x) = x^3 - 4x + c; 8 - 8 + c = 5, \text{ so } c = 5$$

$$f(x) = x^3 - 4x + 5$$

4. $\frac{dy}{dx} = 6x^2 + 2x$ and the curve passes through (-1, 3). Find y in terms of x .

[A]

$$y = 2x^3 + x^2 + c; -2 + 1 + c = 3, \text{ so } c = 4$$

$$y = 2x^3 + x^2 + 4$$

5. $f'(x) = 4x - 1$ and the curve passes through (1, 2). Find $f(3)$.

[A]

$$f(x) = 2x^2 - x + c; 2 - 1 + c = 2, \text{ so } c = 1$$

$$f(3) = 18 - 3 + 1 = 16$$

6. A company's revenue from selling x items is $R = 50x - 0.2x^2$ dollars. Find the rate of change of R when $x = 100$, and explain what it means. [A]

$$\frac{dR}{dx} = 50 - 0.4x; \text{ at } x = 100: 50 - 40 = 10$$

Revenue is increasing at \$10 per extra item sold.

7. The temperature of an oven is $T = 20 + 15t - 0.25t^2$ degrees Celsius after t minutes ($0 \leq t \leq 30$). Find the rate of change at $t = 10$ and explain it. [A]

$$\frac{dT}{dt} = 15 - 0.5t; \text{ at } t = 10: 10$$

The oven is heating up at 10 degrees Celsius per minute.

8. A particle moves with $s = t^3 - 9t^2 + 24t$ metres after t seconds. [A]

- (a) Find its velocity at $t = 1$.

$$v = 3t^2 - 18t + 24; v(1) = 3 - 18 + 24 = 9 \text{ m s}^{-1}$$

- (b) Find the times when it is at rest.

$$3(t^2 - 6t + 8) = 3(t - 2)(t - 4) = 0, \text{ so } t = 2 \text{ s and } t = 4 \text{ s}$$

- (c) Find its acceleration at $t = 4$.

$$a = 6t - 18; a(4) = 6 \text{ m s}^{-2}$$

9. A car has velocity $v = 3t^2 + 2 \text{ m s}^{-1}$ and starts at the origin. How far has it travelled after 4 seconds? [A]

$$s = t^3 + 2t + c; s(0) = 0, \text{ so } c = 0$$

$$s(4) = 64 + 8 = 72 \text{ m}$$

10. A particle starts from rest at the origin with acceleration $a = 6 - 2t \text{ m s}^{-2}$. Find its velocity and displacement at $t = 3$. [A]

$$v = 6t - t^2 \text{ (from rest, } c = 0); v(3) = 18 - 9 = 9 \text{ m s}^{-1}$$

$$s = 3t^2 - \frac{1}{3}t^3 \text{ (from the origin); } s(3) = 27 - 9 = 18 \text{ m}$$

11. The height of a young tree is $H = 0.5 + 0.8t - 0.02t^2$ metres after t years ($0 \leq t \leq 20$). When is it growing at 0.4 m per year? [A]

$$\frac{dH}{dt} = 0.8 - 0.04t = 0.4$$

$$0.04t = 0.4, \text{ so } t = 10 \text{ years}$$

12. A ball is thrown vertically up from a 2 m platform at 12.6 m s^{-1} (acceleration -9.8 m s^{-2}). Find its maximum height above the ground. [M]

$$v = 12.6 - 9.8t; s = 2 + 12.6t - 4.9t^2$$

$$\text{Maximum when } v = 0: t = \frac{12.6}{9.8} = 1.286 \text{ s}$$

$$s = 2 + 12.6(1.286) - 4.9(1.286)^2 = 10.1 \text{ m above the ground}$$

13. A car brakes with velocity $v = 20 - 4t \text{ m s}^{-1}$ until it stops. How far does it travel while braking? [M]

$$\text{Stops when } v = 0: t = 5 \text{ s}$$

$$s = 20t - 2t^2 \text{ (from where braking starts); } s(5) = 100 - 50 = 50 \text{ m}$$

14. **Extension** A particle moves with $v = 3t^2 - 12t + 9 \text{ m s}^{-1}$ and starts at the origin. Find its displacement after 3 seconds, and the total distance it travels in those 3 seconds. [E]

$$s = t^3 - 6t^2 + 9t; s(3) = 27 - 54 + 27 = 0: \text{ displacement } 0 \text{ m}$$

$$\text{At rest when } 3(t-1)(t-3) = 0: \text{ it turns round at } t = 1$$

$$s(1) = 1 - 6 + 9 = 4: \text{ out } 4 \text{ m, then back } 4 \text{ m}$$

$$\text{Total distance} = 4 + 4 = 8 \text{ m}$$

Optional extra: StudyTime Level 2 Calculus Guide, "Quick Questions" on page 29.